CSA5108 Cryptography and Network security

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Exp 10 : Playfair Cipher — Given Matrix Encryption
Code:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

char m[5][5] = {
   {'M','F','H','I','K'},
   {'U','N','O','P','Q'},
   {'Z','V','W','X','Y'},
   {'E','L','A','R','G'},
   {'D','S','T','B','C'}
};

void pos(char c, int *r, int *col) {
   if (c == 'J') c = 'I';
   for (int i=0;i<5;i++)
      for (int j=0;j<5;j++)
         if (m[i][j] == c) {
            *r=i; *col=j; return;
         }
}

int main(void) {
   char in[500], p[500], out[500];
   int n=0, k=0;

   printf("Enter plaintext: ");
   fgets(in,sizeof(in),stdin);

   for(int i=0;in[i];i++)
      if(isalpha((unsigned char)in[i])) {
```

```
        p[n++]=toupper((unsigned char)in[i]);
    }
p[n]='\0';

for(int i=0;i<n;) {
    char a=p[i], b;
    if(i+1>=n) b='X';
    else b=p[i+1];

    out[k++]=a;
    if(a==b) {
        out[k++]='X';
        i++;
    } else {
        out[k++]=b;
        i+=2;
    }
}
if(k%2) out[k++]='X';
out[k]='\0';

for(int i=0;i<k;i+=2) {
    int r1,c1,r2,c2;
    pos(out[i],&r1,&c1);
    pos(out[i+1],&r2,&c2);

    if(r1==r2) {
        out[i]=m[r1][(c1+1)%5];
        out[i+1]=m[r2][(c2+1)%5];
    } else if(c1==c2) {
        out[i]=m[(r1+1)%5][c1];
        out[i+1]=m[(r2+1)%5][c2];
    } else {
        out[i]=m[r1][c2];
        out[i+1]=m[r2][c1];
    }
}
```

```
    printf("Ciphertext: %s\n",out);
    return 0;
}
```

